

Secure Online Voting System Using Cryptography

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA5115 – Cryptography and Network Security

to the award of the degree of

BACHELOR OF ENGINEERING

IN

Computer Science and Engineering

Submitted by

M Nishanth (Reg. No. 192511154)

Under the Supervision of

Dr. T. SANGEETHA



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

September-2026



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105



DECLARATION

We, **M Nishanth** of the **BE Computer Science and Engineering**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “**Secure Online Voting System Using Cryptography**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Students with Names

M Nishanth (192511154)

Saveetha Institute of Medical and Technical Sciences
Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Secure Online Voting System Using Cryptography**” has been carried out by **M Nishanth** under the supervision of **Dr. T. SANGEETHA** and is submitted in partial fulfilment of the requirements for the current semester of the **BE Computer Science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Dr. T. Sangeetha

Professor

Department of

Saveetha School of Engineering

SIMATS

Submitted for the Project work Viva-Voce held on _____

.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. N.M. Veeraiyan, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, Dr. Deepak Nallaswamy Veeraiyan, and our Vice-Chancellor, Dr. S. Suresh Kumar, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, Dr. Ramya Deepak, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, Dr. B. Ramesh for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, **Dr. T. Sangeetha** for his continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide, **Dr. T. Sangeetha** for his creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Students Names

ABSTRACT

Voting is one of the most fundamental mechanisms of a democratic society, and the integrity, confidentiality, and transparency of the voting process are essential to maintaining public trust. Conventional paper-based and even many electronic voting systems suffer from limitations such as booth-capture, human error, delayed counting, and vulnerability to tampering, while lacking mechanisms to verify that a vote has not been altered in transit. With the growing availability of internet connectivity, there is a strong need for a secure online voting system that allows citizens to cast their vote remotely while ensuring that the vote remains confidential, authentic, and tamper-proof. This project focuses on the design and implementation of a Secure Online Voting System Using Cryptography, built around four core modules: Voter Registration, RSA-Based Vote Encryption, Digital Signature Verification, and Vote Counting & Result Analysis. During voter registration, each eligible voter is authenticated and issued a unique cryptographic identity, preventing duplicate or fraudulent registrations. When a voter casts a ballot, the vote is encrypted using the RSA public-key cryptosystem before it leaves the voter's device, ensuring that the content of the vote remains confidential even if intercepted during transmission. A digital signature is then generated and attached to the encrypted vote, allowing the system to verify the authenticity and integrity of each ballot and to detect any attempt at tampering or impersonation. Finally, votes that pass signature verification are decrypted, tallied, and the results are analysed and presented in a transparent and auditable manner. The system is designed using standard cryptographic principles and lightweight software components, making it suitable for academic institutions, small organisations, and pilot-scale democratic elections. The implementation of such a system demonstrates how public-key cryptography and digital signatures can be combined to build a voting platform that is secure, verifiable, and resistant to common forms of electoral fraud.

TABLE OF CONTENTS

S. No.	Content	Page No.
1	Introduction	01
2	Problem Identification and Analysis	03
3	Solution Identification and Analysis	05
4	Results and Recommendations	08
5	Reflection On Learning and Personal Development	10
6	Conclusion	13
7	References	14

CHAPTER 1

INTRODUCTION

1.1 Background Information

Elections form the foundation of democratic governance, and the manner in which votes are collected, protected, and counted directly affects public confidence in the outcome. Traditional paper-based voting requires voters to be physically present at a designated polling booth, which can be inconvenient for people living far from their registered constituency, working professionals, elderly citizens, and those with mobility restrictions. Even where electronic voting machines are used, the process of transmitting and aggregating results still depends heavily on physical custody of hardware and manual verification, leaving room for errors and disputes. In recent years, advances in cryptography and network security have made it possible to design online voting platforms that allow citizens to cast their vote remotely while still preserving the core requirements of an election: voter authentication, ballot confidentiality, vote integrity, and accurate counting. Public-key cryptography, particularly the RSA algorithm, enables a voter's ballot to be encrypted in such a way that only the authorised election authority can decrypt and read it. Digital signatures, generated using the voter's private key, allow the system to confirm that a vote genuinely originated from a registered voter and was not altered after it was cast. The proposed project focuses on developing a Secure Online Voting System Using Cryptography that integrates voter registration, RSA-based vote encryption, digital signature verification, and automated vote counting into a single, cohesive workflow. This approach improves accessibility to the voting process while ensuring that every ballot remains private, authentic, and verifiable.

1.2 Project Objectives

The main objective of this project is to design and implement a secure online voting system that allows registered voters to cast their ballots remotely while guaranteeing confidentiality, authenticity, and integrity of every vote. The project aims to develop a voter registration module that verifies voter eligibility and prevents duplicate registration or impersonation. A further objective is to implement RSA-based encryption so that the content of a ballot cannot be read by any unauthorised party during transmission or storage. The project also aims to incorporate digital signature verification so that any attempt to tamper with a vote or forge a voter's identity can be detected and rejected. Finally, the project aims to build a vote counting and result analysis module that accurately tallies verified votes and presents the results in a transparent and auditable format, demonstrating the practical application of cryptographic

techniques in solving real-world problems of electoral security.

1.3 Significance

Electoral fraud, vote tampering, and low voter turnout due to inaccessibility are significant concerns in modern democratic processes. A secure online voting system addresses these concerns by combining convenience with strong cryptographic guarantees. By encrypting each vote using RSA public-key cryptography and validating it with a digital signature, the system ensures that votes cannot be read, altered, or forged without detection. This significantly reduces the possibility of fraud while making the voting process more accessible to citizens who might otherwise be unable to vote in person. The project also contributes to the broader field of cryptography and network security by demonstrating how established cryptographic primitives can be applied to build trustworthy, real-world systems for sensitive applications such as elections.

1.4 Scope Included

The scope of this project includes the design, development, and testing of a secure online voting platform consisting of four core modules: voter registration, RSA-based vote encryption, digital signature verification, and vote counting with result analysis. The project focuses on demonstrating the cryptographic workflow of an online election, from voter enrolment through to final result declaration, on a small-scale or simulated set of voters. The system does not extend to large-scale national election infrastructure, but is designed to illustrate the core security mechanisms required for trustworthy digital voting.

1.5 Methodology Overview

The system begins with the Voter Registration module, where eligible voters are authenticated and issued a unique digital identity along with a public-private key pair. When a registered voter casts a vote, the RSA-Based Vote Encryption module encrypts the ballot using the election authority's public key, ensuring that the vote's content remains confidential. The Digital Signature Verification module then attaches a digital signature generated with the voter's private key to the encrypted ballot and verifies this signature at the server to confirm the voter's identity and detect tampering. Once verified, votes are securely stored, and the Vote Counting & Result Analysis module decrypts the valid votes using the authority's private key, tallies them, and generates the final results along with an auditable summary of the voting process.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

One of the major challenges in modern elections is ensuring that votes are collected securely, remain confidential, and cannot be altered or forged once cast. Traditional paper-based and many existing electronic voting systems require voters to be physically present, which reduces accessibility and increases logistical costs. When elections are moved online without adequate security measures, they become vulnerable to a range of threats, including vote tampering, voter impersonation, replay attacks, and unauthorised disclosure of how a particular individual voted. Another issue is the lack of verifiable mechanisms to confirm that a counted vote genuinely corresponds to an authenticated, eligible voter and has not been modified after submission. Additionally, manual vote counting is time-consuming and prone to human error, especially in large-scale elections. These challenges highlight the need for an online voting system that uses strong cryptographic techniques to guarantee confidentiality, authenticity, and integrity throughout the voting lifecycle.

2.2 Evidence of the Problem

Electoral integrity is a major concern in both physical and digital voting environments. Traditional systems depend heavily on manual verification and physical security of ballots and machines rather than mathematically provable guarantees of authenticity. This often leads to disputes over vote tampering or miscounted results. Voter impersonation occurs when an unauthorised person casts a vote on behalf of another individual, undermining the fairness of the election. On the other hand, insecure transmission of votes can expose sensitive voting information, violating voter privacy. In many regions, the lack of accessible and trustworthy digital voting options results in lower voter turnout among citizens who are unable to reach polling stations. Studies in the field of cryptography and network security have shown that public-key encryption and digital signatures can significantly improve the confidentiality and authenticity of digital communications. By applying RSA-based encryption and digital signature verification to the voting process, votes can be transmitted and counted securely, helping to preserve both voter privacy and electoral integrity.

2.3 Stakeholders

The secure online voting system affects several stakeholders in the electoral process. The primary stakeholders are voters, who require a convenient and trustworthy way to exercise their right to vote without compromising the confidentiality of their choice. Election authorities and administrators are also key stakeholders, as they are responsible for verifying voter eligibility, managing cryptographic keys, and ensuring that the final results accurately reflect the votes cast. Another group of stakeholders includes auditors and observers who need to verify that the election was conducted fairly and that no votes were tampered with or fraudulently added. Government bodies and regulatory agencies are important stakeholders as well, since they are responsible for certifying that digital voting systems meet the required legal and security standards before being adopted for real elections.

2.4 Supporting Data/Research

Modern network security research emphasises the importance of confidentiality, integrity, and authentication in any system that transmits sensitive data over a network. Public-key cryptosystems such as RSA play a central role in securing digital communication by allowing data to be encrypted with a public key and decrypted only with the corresponding private key. Digital signature schemes, which are also based on public-key cryptography, allow a receiver to verify both the identity of the sender and that a message has not been altered in transit. These techniques are widely used in secure applications such as online banking, digital contracts, and secure messaging, and are equally applicable to online voting. Combining RSA-based encryption with digital signature verification allows an online voting system to keep the content of each vote confidential while still allowing the system to confirm that the vote came from an authenticated, eligible voter, without revealing which candidate the voter selected. This combination of cryptographic techniques forms the technical foundation for building a trustworthy and tamper-resistant online voting platform.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development of the secure online voting system began with identifying the main problem in existing voting mechanisms, which is the absence of strong cryptographic guarantees for confidentiality, authenticity, and integrity in remote voting. After understanding the problem, a modular solution was designed consisting of four cooperating components: Voter Registration, RSA-Based Vote Encryption, Digital Signature Verification, and Vote Counting & Result Analysis. The design process involved selecting appropriate cryptographic algorithms that could securely protect a vote from the moment it is cast until it is counted. RSA was chosen as the encryption scheme because it is a well-established public-key cryptosystem suitable for encrypting small data items such as a ballot selection, and because the same underlying key pairs can be reused for generating and verifying digital signatures. The development process also included testing the system with simulated voters and sample elections to confirm that registration, encryption, signature verification, and tallying worked correctly together. After successful testing, the modules were integrated to form a complete working model of the secure online voting system.

3.2 Tools and Technologies Used

Several software tools and cryptographic libraries were used in the development of this project. The core cryptographic operations, including RSA key generation, encryption, decryption, and digital signature creation and verification, were implemented using standard cryptography libraries available in languages such as Python and Java. A simple web-based or command-line interface was used to allow voters to register and submit their encrypted, signed ballots. A lightweight database was used to securely store voter records, public keys, and encrypted votes until the counting stage. Hashing algorithms such as SHA-256 were used alongside RSA to generate compact message digests for the digital signature process, ensuring that even large ballots could be signed efficiently. These tools and technologies together helped create a secure, modular, and testable online voting platform.

3.3 Solution Overview

The proposed solution is a secure online voting system built around four integrated modules.

3.3.1 Module 1 – Voter Registration

In this module, each citizen who wishes to vote is registered and their eligibility is verified against the electoral roll. During registration, the system generates a unique RSA public-private key pair for the voter. The private key is securely delivered to the voter, while the public key is stored by the election authority and used later to verify the voter's digital signature. This module also prevents duplicate registrations and ensures that only authenticated, eligible voters can proceed to cast a vote.

3.3.2 Module 2 – RSA-Based Vote Encryption

When a registered voter casts their ballot, the vote is encrypted using the RSA public key of the election authority before it is transmitted over the network. This ensures that the content of the ballot, that is, the candidate selected by the voter, remains confidential and unreadable to anyone who might intercept the transmission, including the system administrators, until it is officially decrypted during the counting stage. Only the election authority's private key can decrypt the vote, which keeps the ballot secure throughout its journey from the voter's device to the central server.

3.3.3 Module 3 – Digital Signature Verification

Alongside encryption, each vote is digitally signed using the voter's private key generated during registration. The system computes a hash of the encrypted ballot and signs it, producing a digital signature that is transmitted along with the vote. On receiving the vote, the server uses the voter's stored public key to verify the signature. If the signature is valid, the system confirms that the vote genuinely originated from the registered voter and has not been altered in transit. If the signature verification fails, the vote is flagged and rejected, preventing tampered or forged ballots from being counted.

3.3.4 Module 4 – Vote Counting & Result Analysis

Once all votes have been collected and their digital signatures verified, the election authority uses its private key to decrypt each valid, verified vote. The decrypted votes are then tallied automatically by the system, and the results are compiled into a summary showing the number of votes received by each candidate. The module also generates an auditable log of the counting process, allowing observers to confirm that only verified votes were included in the final result, without exposing which specific voter

selected which candidate.

3.4 Engineering Standards Applied

During the design and implementation of this project, standard cryptographic and network security principles were followed to ensure reliable operation of the system. Adequate RSA key lengths were used to maintain a strong level of security against brute-force attacks. Secure hashing algorithms were applied prior to digital signature generation to protect against forgery. Standard secure coding practices were followed while implementing the encryption, signing, and verification routines to avoid common vulnerabilities. The project also follows the principle of separation of duties, keeping voter identity verification, vote encryption, and vote counting as distinct modules, which improves the overall security and auditability of the system.

3.4 Solution Justification

The proposed solution is justified because it directly addresses the core problems of confidentiality, authenticity, and integrity in online voting. Traditional and unsecured digital voting methods often risk exposing voter choices or allowing tampered votes to be counted. By using RSA-based encryption, the system ensures that vote content remains private, while digital signature verification ensures that only genuine, unaltered votes cast by eligible voters are accepted. The modular design, separating registration, encryption, verification, and counting, makes the system easier to test, audit, and extend. Therefore, the solution provides strong security guarantees while remaining practical to implement for small-to-medium scale elections.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The developed secure online voting system was tested to evaluate its performance and effectiveness across all four modules. During testing, the voter registration module successfully verified voter eligibility and issued unique key pairs without allowing duplicate registrations. The RSA-based vote encryption module correctly encrypted sample ballots such that the vote content could not be read without the election authority's private key. The digital signature verification module was able to correctly accept votes signed with valid voter keys and reject votes that were deliberately tampered with or signed using an incorrect key, demonstrating that forged or altered ballots could be reliably detected. The vote counting and result analysis module accurately decrypted and tallied all verified votes, producing a correct summary of results for the simulated election. Overall, the results indicate that the system is capable of providing a secure, private, and verifiable online voting experience.

4.2 Challenges Encountered

During the implementation of the project, several challenges were encountered. One of the main challenges was managing RSA key generation and secure key distribution, since the private key of each voter had to be delivered securely without exposing it to unauthorised parties. Another challenge was ensuring that the digital signature verification process correctly identified genuine votes while rejecting tampered ones, which required careful implementation of the hashing and signing routines. Performance was also a consideration, since RSA operations can be computationally expensive when applied to a large number of voters; this was addressed by encrypting only the compact ballot data and using hashing before signing. Despite these challenges, the system was successfully implemented and tested on a simulated set of voters.

4.3 Possible Improvements

Although the current system performs its intended function, several improvements can be made in future versions of the project. One possible improvement is the integration of blockchain technology to provide an immutable, publicly verifiable record of encrypted votes. Another improvement could be the use of hybrid encryption, combining RSA with a symmetric algorithm such as AES, to improve

performance for large-scale elections. Multi-factor authentication could also be added to the voter registration module to further strengthen voter identity verification. Additionally, a real-time dashboard could be developed to allow election observers to monitor the voting and counting process transparently.

4.4 Recommendations

It is recommended that future research focus on developing more advanced secure voting systems that incorporate blockchain-based ledgers, hybrid encryption schemes, and zero-knowledge proof techniques to further enhance voter privacy and result verifiability. Additionally, further testing can be conducted with larger simulated voter populations and under adversarial conditions to evaluate the system's resilience against attacks such as replay attacks and denial-of-service attempts. By implementing these improvements, the secure online voting system can become more efficient and suitable for larger-scale real-world elections.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

5.1.1 Academic Knowledge

Through the development of the project titled “Secure Online Voting System Using Cryptography,” I gained significant academic knowledge related to public-key cryptography, digital signatures, and network security principles. This project allowed me to apply theoretical concepts that I learned in my coursework, such as the RSA algorithm, hashing functions, and secure key management. One of the important concepts I learned was how public and private key pairs work together to provide both confidentiality through encryption and authenticity through digital signatures. I also learned how cryptographic systems can be used to solve real-world trust problems, such as ensuring that an online vote is both private and verifiable. Additionally, the project enhanced my understanding of secure system design, where multiple modules, registration, encryption, signature verification, and counting, must work together without compromising overall security. This helped me understand how technology can be used to solve real-world problems such as electoral fraud and lack of trust in digital systems. Overall, the project strengthened my knowledge in cryptography, secure programming, and system design, and it showed how theoretical knowledge can be applied practically to create a trustworthy engineering solution.

5.1.2 Technical Skills

This project helped me develop several important technical skills related to both cryptographic implementation and software development. One of the key skills I gained was implementing RSA key generation, encryption, and decryption using standard cryptography libraries. I learned how to generate secure key pairs, encrypt and decrypt data, and manage cryptographic keys safely. I also gained hands-on experience with digital signature schemes, including hashing a message and signing it with a private key, and verifying that signature with the corresponding public key. I learned how to design a modular system architecture, connecting the voter registration, encryption, signature verification, and vote counting components into a single working pipeline.

Another important technical skill I developed was troubleshooting and debugging cryptographic code. During the implementation process, I encountered issues such as signature verification failures caused by incorrect key handling and encoding mismatches between the encryption and decryption stages. By carefully analysing the cryptographic workflow and testing each module independently, I was able to

identify and fix these issues. Furthermore, I improved my skills in system integration, which involved combining multiple security-focused components to work together as a complete, coherent voting platform. These technical skills are highly valuable for future projects and professional work in the fields of cryptography, network security, and secure software engineering.

5.1.3 Problem Solving and Critical Thinking

The project also helped me improve my problem-solving and critical thinking abilities. During the development process, several technical challenges were encountered that required careful analysis and creative solutions. One of the major problems faced was ensuring that RSA encryption and digital signatures worked correctly together without conflicting, since both operations rely on the same underlying key pairs but serve different security purposes. To overcome this issue, I studied the mathematical principles behind RSA and carefully separated the encryption and signing logic in the implementation. Another challenge was designing the system so that a tampered or forged vote would always be detected and rejected, which required rigorous testing with intentionally corrupted data. These challenges encouraged me to think critically and apply logical reasoning when solving technical problems. Instead of relying only on theoretical knowledge, I learned how to experiment, analyse results, and refine the system to achieve the desired level of security. As a result, this project significantly improved my confidence in solving engineering problems and strengthened my ability to approach technical challenges in a systematic and effective manner.

5.2 Challenges Encountered and Overcome

5.2.1 Personal and Professional Growth

During the development of this project, I faced several challenges that contributed to my personal and professional growth. One of the main difficulties was designing a system that could securely encrypt, sign, and verify votes without introducing vulnerabilities that could be exploited by an attacker.

Initially, the digital signature verification process occasionally failed due to inconsistencies in how data was hashed before signing and verifying. This made it difficult to reliably distinguish genuine votes from tampered ones. To overcome this problem, I conducted several tests using controlled, intentionally altered data and carefully traced the cause of each verification failure. Another challenge was ensuring that private keys generated during voter registration were handled securely and not accidentally exposed in logs or intermediate files. Through repeated testing and careful code review, I was able to identify the issues and

correct them. These challenges helped me develop patience, persistence, and determination. I learned that security-focused engineering projects often require multiple trials and improvements before achieving a successful, reliable result. Overcoming these challenges strengthened my problem-solving abilities and increased my confidence in handling complex technical tasks.

5.2.2 Collaboration and Communication

Collaboration and communication played an important role during the development of this project. Throughout the project, I interacted with classmates and project supervisors to discuss ideas, clarify doubts, and improve the design of the voting system. These discussions helped me understand the importance of teamwork in engineering projects. By sharing ideas and receiving feedback, I was able to improve both the cryptographic design and the overall system architecture used in the project. Communication with my project guide also helped me gain valuable insights and suggestions for improving the security of the voting system. For example, suggestions were given regarding key management practices and testing methods for detecting tampered votes. Working collaboratively also improved my ability to explain technical cryptographic concepts clearly and listen to others' suggestions. These communication and teamwork skills are essential in professional environments where engineers often work together to design and develop complex, security-critical systems.

5.3 Application of Engineering Standards

During the design and implementation of the secure online voting system, basic engineering standards and cryptographic best practices were followed to ensure reliability and security. Proper key length selection was carried out to ensure that the RSA implementation remained secure against brute-force attacks while still functioning efficiently. Security considerations were taken into account when handling private keys, encrypted votes, and digital signatures, including secure storage and careful separation of duties between modules. The system was designed in a modular way so that individual components such as registration, encryption, signature verification, and vote counting could be tested separately before integrating them into the complete system. This approach follows standard engineering practices used in secure system design and testing. Additionally, the project emphasised the use of well-established, peer-reviewed cryptographic algorithms rather than custom or untested schemes. By following these engineering principles and best practices, the project achieved a reliable and practical solution for secure online voting.

5.4 Insights into the Industry

This project provided valuable insights into how modern cryptographic technology is being used to secure sensitive digital processes such as elections. Secure electronic and online voting systems are becoming increasingly important due to the growing demand for accessible, trustworthy democratic participation. Traditional voting methods often require physical presence and manual processes, while insecure digital alternatives risk exposing votes or allowing undetected tampering. In contrast, systems built on public-key cryptography and digital signatures can provide strong guarantees of confidentiality and authenticity even over untrusted networks. Through this project, I learned how cryptographic primitives such as RSA and digital signatures are used in real-world security-critical applications, extending beyond voting to areas such as online banking, secure messaging, and digital identity verification. This experience helped me understand the importance of integrating strong security principles into the design of any system that handles sensitive data. The project also highlighted the growing demand for secure, verifiable digital systems that can be trusted by the public. This experience has increased my interest in developing secure systems and exploring advanced cryptographic techniques such as blockchain and zero-knowledge proofs that can contribute to trustworthy digital governance.

5.5 Conclusion of Personal Development

The completion of this capstone project has contributed significantly to my overall personal and professional development. It provided an opportunity to apply theoretical cryptography knowledge in a practical environment and develop a functional, security-focused engineering solution. Through the project, I improved my technical skills in cryptography, secure programming, and system design. I also developed important soft skills such as problem-solving, critical thinking, communication, and teamwork. The experience of designing and implementing a secure online voting system has increased my confidence in handling real-world engineering challenges related to security. It has also motivated me to explore more advanced technologies related to applied cryptography, blockchain, and secure distributed systems, this project has played an important role in preparing me for future career opportunities in the field of cryptography and network security.

CHAPTER 6

CONCLUSION

The project titled “Secure Online Voting System Using Cryptography” was successfully designed and implemented to address the common problem of ensuring confidentiality, authenticity, and integrity in remote voting. Elections form the foundation of democratic governance, and trustworthy vote collection and counting are essential for maintaining public confidence in the outcome. However, many traditional and existing digital voting methods either require physical presence or lack strong cryptographic guarantees, which can lead to reduced accessibility, voter impersonation, or undetected tampering. The developed system aims to overcome these issues by introducing a secure online voting mechanism that protects each vote from the moment it is cast to the moment it is counted.

The main objective of this project was to develop a simple, reliable, and cryptographically secure voting system built around four cooperating modules. The Voter Registration module verifies voter eligibility and issues each voter a unique RSA key pair. The RSA-Based Vote Encryption module ensures that the content of a ballot remains confidential by encrypting it with the election authority's public key before transmission. The Digital Signature Verification module confirms that each vote genuinely originates from a registered voter and has not been tampered with, by verifying a digital signature generated using the voter's private key. Finally, the Vote Counting & Result Analysis module decrypts and tallies only the votes that pass signature verification, producing an accurate and auditable set of results.

One of the key advantages of this system is its ability to protect voter privacy while still allowing the election authority to confirm the authenticity of every vote counted. This not only safeguards the confidentiality of individual voting choices but also ensures that only genuine, unaltered votes contribute to the final result. In addition, allowing voters to cast their ballots remotely reduces the logistical burden of physical polling stations and improves accessibility for voters who may otherwise be unable to participate. Another important aspect of the project is that it is built using well-established, standard cryptographic algorithms such as RSA and secure hashing, which are widely studied, publicly reviewed, and considered reliable for security-critical applications. This makes the system both practical to implement and reasonably trustworthy for small-to-medium scale elections.

During the development of this project, several practical skills and technical knowledge were gained. The project provided hands-on experience in working with public-key cryptography, digital signature schemes, secure key management, and modular system design. It also improved problem-solving abilities

and helped in understanding how cryptographic systems can be used to solve real-world challenges related to trust and security. Through testing and implementation, the importance of careful key handling, consistent hashing before signing, and reliable verification logic was also learned. Furthermore, this project highlights the importance of integrating modern cryptographic technology into critical civic processes such as elections to improve accessibility, transparency, and trust. With increasing concerns about electoral fraud and the need for accessible digital participation, secure online voting systems like the one developed in this project can play a significant role in supporting trustworthy digital governance. In conclusion, the secure online voting system based on RSA encryption and digital signature verification successfully demonstrates how established cryptographic techniques can be used to improve the confidentiality, authenticity, and integrity of the voting process. The system provides an effective solution for reducing electoral fraud, protecting voter privacy, and ensuring accurate vote counting. With further improvements such as blockchain-based ledgers or hybrid encryption schemes, the system can be expanded into a more advanced, large-scale secure voting solution for modern digital democracy.

CHAPTER 7

REFERENCES

1. Reddy, C. S., Chaitra, M., Aishwarya, P. G., & Pravalika, P. (2026, January). Visual Cryptography-Enabled Voting System. In *2026 IEEE International Conference on Emerging Computing and Intelligent Technologies (ICoECIT)* (pp. 1-4). IEEE.
2. Kavitha, R., Meera, S. N., Saibhargav, C., & Niranjani, S. The Comprehensive Online Voting System for Digital Democracy.
3. Pavithra, H. B., Chinmayi, G. S., Pruthvi Acharya, J., & KS, S. (2026, April). Cryptography-Based Secure Voting System for College Elections. In *2026 4th International Conference on Knowledge Engineering and Communication Systems (ICKECS)* (pp. 1-5). IEEE.
4. Deepa, K., Anuvarshini, S. B., Girinandhini, K. S., Naveena, K., Velmani, B., & Kannadhasan, S. (2026, January). Digital Safe Voting using Hybrid Cryptography and Steganography with Deep Learning-based Authentication. In *2026 International Conference on AI-Driven Smart Systems and Ubiquitous Computing (ICAUC)* (pp. 1698-1705). IEEE.
5. Nithya, G., Kiruthika, D., Dharshini, S., & Devi, K. G. (2026, May). A Secure Decentralized Online Voting System Using Blockchain with SHA-512-Based Voter Authentication and Tamper-Proof Vote Validation. In *2026 IEEE Nexo-Tech-International Conference on Advanced Technologies and Innovations (ICATI)* (pp. 1-7). IEEE.
6. Yadav, N. K., Meenakshi, K., & Kushwaha, P. K. (2026, February). Design and Implementation of a Secure Blockchain-Enabled Online Voting System Using Multi-Factor Authentication. In *2026 IEEE 4th International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC)* (pp. 223-227). IEEE.
7. Balamurugan, K., Suganya, S., Pavithra, S., Gokulnath, S., Jagathisan, R., & Rasika, S. K. (2026, May). Enhanced Security and Efficiency of Online Voting Systems Using the SEDP Algorithm in Blockchain: A Comparison with Traditional Systems. In *2026 IEEE Nexo-Tech-International Conference on Advanced Technologies and Innovations (ICATI)* (pp. 1-5). IEEE.
8. Neelammai, J. P., Keerthana, S., Aarthi, R., & Hamshika, S. B. (2026). Blockchain-Based online voting system using PoA consensus. *Mathematics and Computational Sciences*, 7(1), 34-44.
9. Nwamouh, C. U., Aru-Eze, O., Ahiara, W. C., & Samuel, I. A. (2027). Cryptographic Integrity and Scalability Challenges in Ethereum-Based E-Voting Systems: Current State of the Art. *Emerging Trends in Sustainable Energy, Smart Systems, and Green Technology*, 741-757.

